

1(i)

Given:

$$f(x) = ((x^2 + 1) / (x^2 - 1))^{1/3}$$

$$u(x) = x^2$$

$$v(t) = (t + 1) / (t - 1)$$

$$w(s) = s^{1/3}$$

Then:

$$f(x) = w(v(u(x)))$$

1(ii)

$$x^2 - 1 \neq 0$$

$$x \neq -1, 1$$

Domain:

$$(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

Range:

Let $t = x^2$, so $t \geq 0$ and $t \neq 1$.

$$R = (t + 1) / (t - 1)$$

As t varies over $[0, \infty)$ except 1, R takes values:

$$(-\infty, -1] \cup (1, \infty)$$

Then $f(x) = \text{cube_root}(R)$. Cube root doesn't change which values are possible, just cube roots them, so the range is:

$$(-\infty, -1] \cup (1, \infty)$$

2(i)

$\cos(\pi x)$ is between -1 and 1

$\cos(\pi x) - 1$ is between -2 and 0

$|\cos(\pi x) - 1|$ is between 0 and 2

$2 - 4|\cos(\pi x) - 1|$ is between 2 - 8 and 2

Range:

$$[-6, 2]$$

2(ii)

$$g(x) = \cos(x)$$

$\cos(\pi x)$ is between -1 and 1.

So $\cos(\pi x) + 1$ is between 0 and 2, so it's never negative

So $|\cos(\pi x) + 1| = \cos(\pi x) + 1$.

Then:

$$f(x) = 2 - 4(\cos(\pi x) + 1)$$

$$f(x) = -4\cos(\pi x) - 2$$

so:

Amplitude: 4

Midline: $y = -2$

Period: 2

Range: $[-6, 2]$

My six plots:

1. Plot 1: $g(x) = \cos(x)$
2. Plot 2: $g_1(x) = \cos(\pi x)$
period becomes 2
3. Plot 3: $g_2(x) = \cos(\pi x) + 1$
midline becomes $y = 1$
4. Plot 4: $g_3(x) = |\cos(\pi x) + 1|$
it's always ≥ 0
5. Plot 5: $g_4(x) = -4|\cos(\pi x) + 1|$
6. Plot 6: $g_5(x) = 2 - 4|\cos(\pi x) + 1|$
shift up by 2

3(i)

$$P(t) = P(0) * 2^{(t/5)}$$

3(ii)

$$10000 = P(0) * 2^{(20/5)}$$

$$10000 = P(0) * 16$$

$$P(0) = 625$$

4(i)

$$\log_4(48) - \log_4(3) = \log_4(48/3) = \log_4(16) = 2$$

4(ii)

$$10^{-(1/2) \log_{10}(49)}$$

$$= (10^{\log_{10}(49)})^{-1/2}$$

$$= 49^{-1/2}$$

$$= 1/7$$

4(iii)

$$\log_a(b) = \ln(b)/\ln(a)$$

$$\log_b(a) = \ln(a)/\ln(b)$$

$$1 / \log_b(a) = 1 / (\ln(a)/\ln(b)) = \ln(b)/\ln(a) = \log_a(b)$$

4(iv)

$$\arcsin(-1/2) = -\pi/6$$

$$\tan(-\pi/6) = -1/\sqrt{3} = -\sqrt{3}/3$$

4(v)

$$\text{Let } f(x) = (1 - e^x)/(1 + e^x)$$

$$f(-x) = (1 - e^{-x})/(1 + e^{-x})$$

Multiply numerator and denominator by e^x :

$$f(-x) = (e^x - 1)/(e^x + 1)$$

$$= -(1 - e^x)/(1 + e^x)$$

$$= -f(x)$$

so f is odd